

Rahat Imroze Ahmed

Chottogram, Bangladesh • +880-01857850139

• rahatahmed171@gmail.com • <https://www.linkedin.com/in/rahat-imroze-bb534819a/> • <https://github.com/RahatAhmed171>

EDUCATION

BSc. in Computer Science and Engineering (2022-2026)

Premier University Chottogram, CGPA: 3.53/4.00

- Thesis: "Bengali Handwritten Subdistrict (Upazila)-District Pair Name Recognition using Ensemble Deep Learning."
- Relevant coursework: Machine Learning, Deep Learning, Computer Networks, Linear Algebra, Algorithms, Databases

RESEARCH EXPERIENCE

Undergraduate Researcher – Computer Vision and Deep Learning Thesis (2025-2026)

Premier University Chottogram, Department of Computer Science and Engineering, Supervisor: Mohammad Hasan

- **Curated a novel dataset** of 9,808 segmented images of Bengali Handwritten Subdistrict-District pair name across 496 unique classes by designing an unsupervised text segmentation pipeline using MSER and Canny edge detection.
- **Proposed a state-of-the-art ensemble** (ViT-Base + ConvNeXt-Tiny) via weighted soft voting, achieving a **94.65% test accuracy**.
- **Submitted findings to a peer-reviewed conference.** ([Paper Link](#))

PROFESSIONAL EXPERIENCE

Dreamers25 Consultancy, Student Visa Counselor, November 2025 – Present

- Guide students in selecting international universities and programs that align with their career goals and academic backgrounds.
- Oversee the end-to-end visa application process, ensuring all financial and academic documents meet strict embassy requirements.

DozentLMS, Technical Writer, November 2022 – February 2023 (Contract)

- Creating comprehensive user manuals, API guides, and installation documentation for specialized WordPress LMS plugins.
- Develop step-by-step tutorials for course creation and student management tools to enhance the overall user experience.

PROJECTS

Sky Image-Based AQI (Air Quality Index) Classification System

- **Developed an end-to-end computer vision pipeline** using to classify sky images into distinct Air Quality Index (AQI) categories based on atmospheric visibility and haze levels. ([Project Link](#))
- Technologies Used: Python, Pytorch, OpenCV, TensorFlow, Pandas, Huggingface

LLM-Powered Financial Tracker Bot

- **Integrated the Gemini API** to parse natural language income and expense inputs into structured database transactions for automated financial logging and real-time summaries. ([Project Link](#))
- Technologies Used: Laravel, MySQL, Webhooks, Gemini API

SKILLS

- **Programming Languages:** Python, JavaScript, PHP, SQL
- **Machine Learning, Deep Learning Libraries:** PyTorch, TensorFlow, OpenCV, Langgraph
- **Frameworks and Technologies:** Laravel, NodeJS, FastAPI, MySQL, Git, Docker
- **Soft Skills:** Problem Solving and Team Work

ACTIVITIES & ACHIEVEMENTS

- Participated in a team-based contest, solving advanced algorithmic problems using C++/Python (CSE IT FEST 2024 PUC)
- General Members at Chittagong Astronomical Society